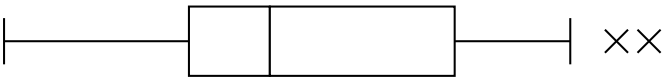
Representations of data

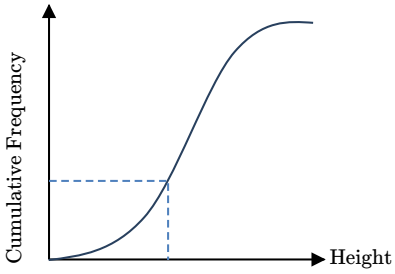
We've seen so far how data is collected and calculations can be made. We now concentrate on how the processed data can be *displayed*.

BOX PLOTS AND OUTLIERS

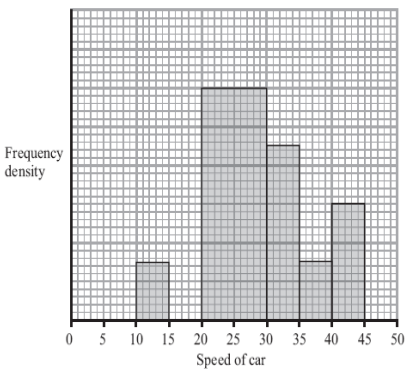


NEW since GCSE! Outliers.

CUMULATIVE FREQ DIAGRAMS



HISTOGRAMS



NEW since GCSE!
Area is not necessarily equal to frequency.
Forming a frequency polygon by joining midpoints.

Changes since the old S1 syllabus

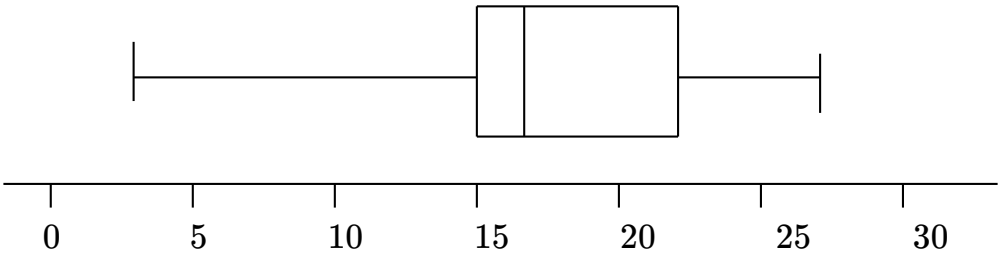
- Stem-and-leaf diagrams and skewness were removed.
- Cumulative frequency diagrams were added.
- Draw a frequency polygon from a histogram.

Box Plot recap

Luisa · 3

Box Plots allow us to visually represent the distribution of the data.

Minimum	Lower Quartile	Median	Upper Quartile	Maximum
3	15	17	22	27

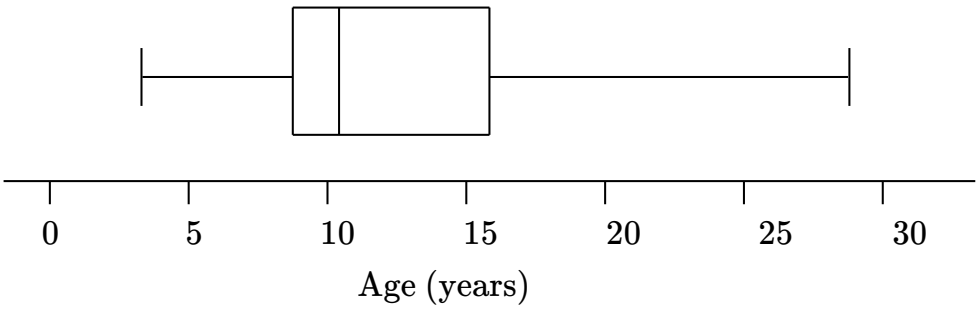


How is the **IQR** represented in this diagram?

How is the **range** represented in this diagram?

Interpreting a Box Plot

Luisa · 4



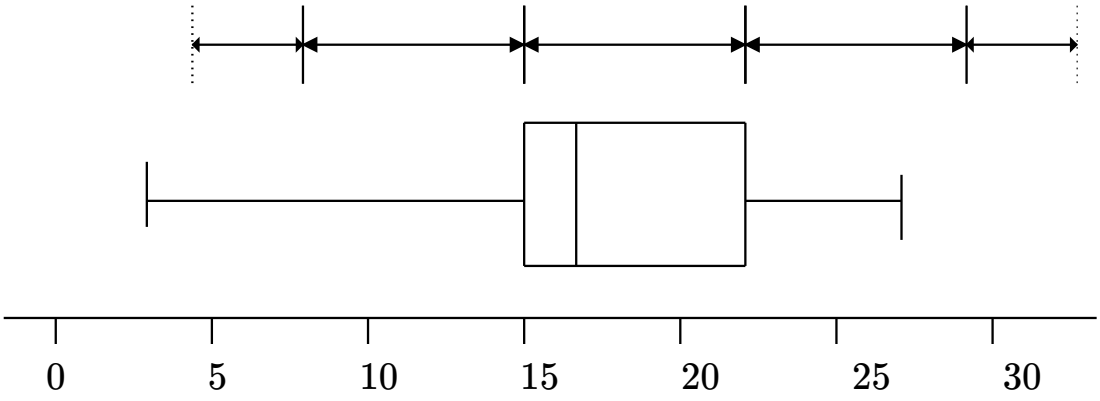
Decide whether each statement is true or false. Explain your reasoning.

"The right box represents more people than the left box."

"The ages are more spread out above the median."

An outlier is

Outliers beyond this point



One common definition of an outlier is when we're **1.5 IQRs** beyond the lower and upper quartiles.

(But you will be told in the exam if the rule differs from this)

The diameters of 11 different Roman coins are measured in centimetres:

2.2 2.5 2.7 2.7 2.8 3.0 3.1 3.1 3.2 4.0 4.7

Determine the quartiles and hence any outliers.

[Textbook] The lengths, in cm, of 12 giant African land snails are given below:

17 18 18 19 20 20 20 20 21 23 24 32

a) Calculate the mean and standard deviation, given that $\Sigma x = 252$ and $\Sigma x^2 = 5468$.

b) An outlier is an observation which lies ± 2 standard deviations from the mean. Identify any outliers for this data.

a

b

Context: Recall that the standard deviation is, roughly speaking, the average distance of each value from the mean. So the outlier definition is saying we're at least twice this average distance, which seems like a sensible definition.

In Year 2, you will encounter the **normal distribution**, which can be used to model data which is **clustered about some mean and tails off symmetrical in either direction**. If this data was approximately normally distributed, then there is a 5% chance a random observation would fall outside 2 standard deviations within the mean. You will learn then how to make such probability calculations.

Test Your Understanding — Luisa · 7

The ages of 15 Lib Dem MPs are given:

11 18 20 27 30 31 32 32 35 36 37 58 63 78 105

a) If an outlier is considered to be 1.5 interquartile ranges below the lower quartile or above the upper quartile, determine any outliers.

b) If instead an outlier is considered to be outside 2 standard deviations within the mean, determine any outliers. Note that $\Sigma x = 613$ and $\Sigma x^2 = 33815$

a

b

Box Plot Example — Luisa · 8

Smallest values	Largest values	Lower Quartile	Median	Upper Quartile
0, 3	21, 27	8	10	14

Draw a box plot to represent the above data.

Exam Tip: You MUST show your outlier boundary calculations.

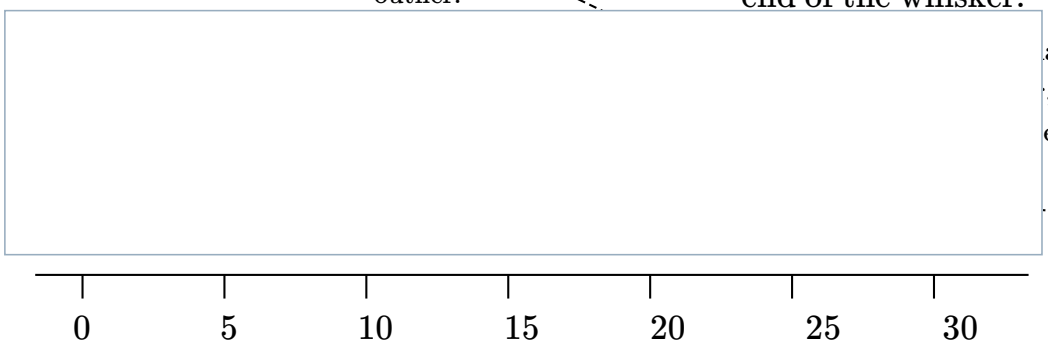
When there's an outlier at one end, there's two allowable places to put the end of the whisker:

Use a cross for each outlier.

aximum value not an
, 21 (I think this one makes
ense).

OR the outlier boundary, 23.

Use one or the other **(not both)**.



Test Your Understanding — Luisa · 9

Jan 2011 Q3. A small company recorded monthly sales over a long period. An outlier is more than 1.5 interquartile ranges beyond either quartile.

(a) Draw a box plot on graph paper indicating outliers. (5 marks)

(c) The company claims that sales exceed £10,000 in 75% of months. Comment on this claim. (2 marks)

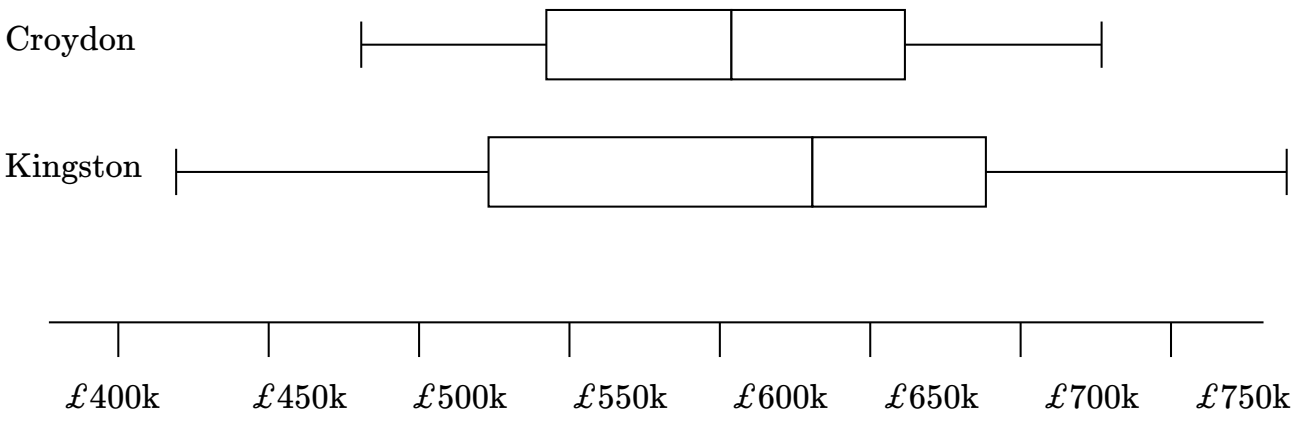
a

Summary	Sales (£1000s)
Two lowest	3, 4
Lower quartile	7
Median	12
Upper quartile	14
Two highest	20, 25

c

Comparing Box Plots — Luisa · 10

Box Plot comparing house prices of Croydon and Kingston-upon-Thames:



"Compare the prices of houses in Croydon with those in Kingston". (2 marks)

For 1 mark, one of:

For 1 mark:

Exercise 3A – Statistics Year 1 Pages 43 **Luisa · 11**

4. The ages of nine people at a children's party have $\sum x = 92, \sum x^2 = 1428$.

(a) Calculate the mean and standard deviation. (3 marks)

An outlier lies more than two standard deviations from the mean. One age is 30.

(b) Is 30 an outlier? Give a reason. (2 marks)

(c) Suggest why it could be legitimate. (1 mark)

(d) Given that all nine were children, clean the data and recalculate the mean and standard deviation. (3 marks)

Hint: find new values of n, $\sum x$ and $\sum x^2$.

a

b

c

d

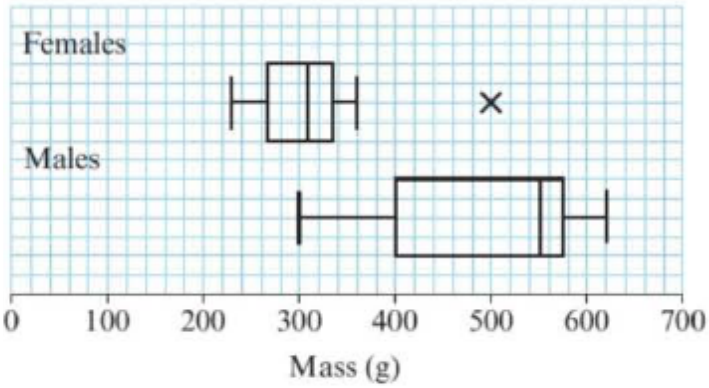
Exercise 3B – Statistics Year 1 Pages 45 **Luisa · 12**

3. The box plots show male and female turtle masses in grams.

(a) Compare their masses.

(b) A turtle weighs 330 g. Is it more likely male or female? Give a reason.

(c) Write down the mass of the largest female.



a

b

c

4. The table gives daily maximum gusts (knots) in Camborne in September 1987 from the large data set. © Crown Copyright Met Office.

(a) Calculate Q_1, Q_2, Q_3. (3 marks)

(b) Using fences 1.5 IQRs beyond the quartiles, show that 46 and 78 are outliers. (1 mark)

(c) Draw a box plot. (3 marks)

13	17	19	20	21
21	22	23	24	25
25	25	26	26	26
27	29	30	30	30
33	35	38	46	78

a

b

c

Supplementary Questions: **Luisa · 14**

5. May 2006 Q1. Children from schools A and B run for charity. Times are recorded to the nearest minute. Figure 1 summarises school A.

(a) Describe the features and uses of a box plot. (3 marks)

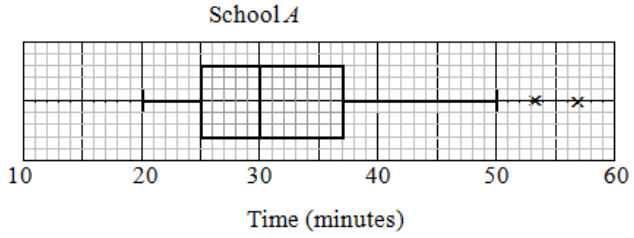
(b) (i) By what time had 75% of A finished? (ii) Name this value. (2 marks)

(c) Explain the two crosses. (2 marks)

School B had minimum 25, quartiles 30, 37 and 50, and maximum 55 minutes.

(d) Draw a box plot for B. (4 marks)

(e) Compare the two plots. (4 marks)



a

b

c

Solutions to (d) and (e) follow.

5. May 2006 Q1. Children from schools A and B run for charity. Times are recorded to the nearest minute. Figure 1 summarises school A.

(a) Describe the features and uses of a box plot. (3 marks)

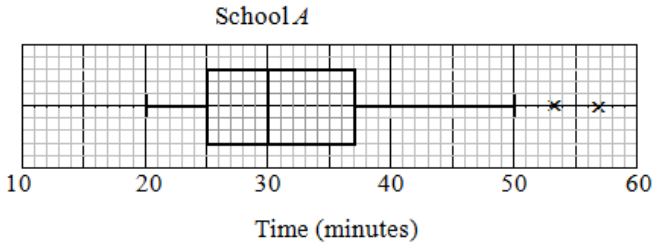
(b) (i) By what time had 75% of A finished? (ii) Name this value. (2 marks)

(c) Explain the two crosses. (2 marks)

School B had minimum 25, quartiles 30, 37 and 50, and maximum 55 minutes.

(d) Draw a box plot for B. (4 marks)

(e) Compare the two plots. (4 marks)



d

e

6. June 2005 Q4. Take-off delays from City A to B had minimum 5 and maximum 63 minutes. A quarter were at most 12 minutes, half at most 17 and three quarters at most 28. Only one exceeded 45 minutes. Use fences 1.5 IQRs beyond the quartiles.

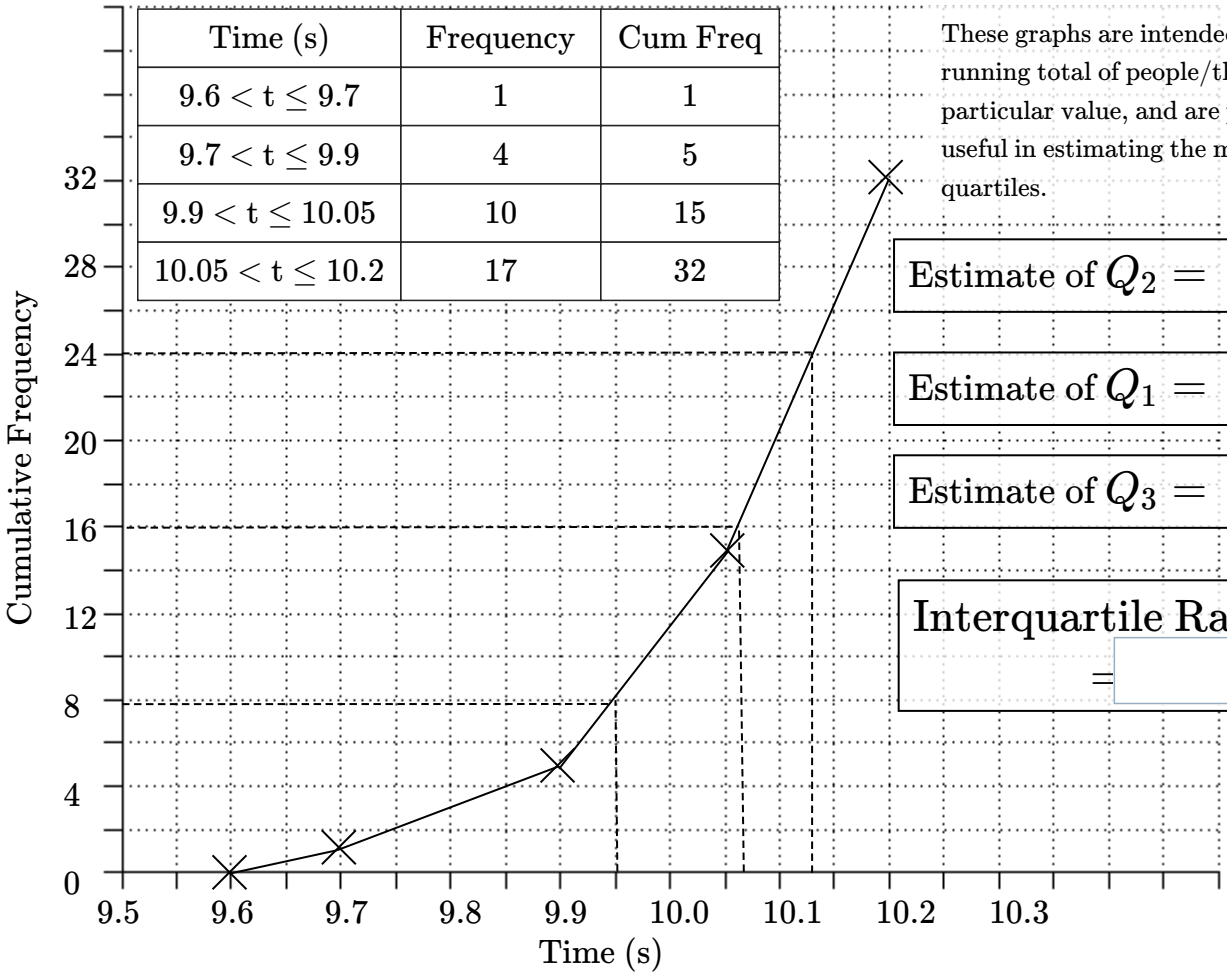
(a) Draw a box plot. (7 marks) (b) Comment on the distribution and justify. (2 marks) (c) Interpret it for a frequent passenger. (1 mark)

a

b

c

Time (s)	Frequency	Cum Freq
$9.6 < t \leq 9.7$	1	1
$9.7 < t \leq 9.9$	4	5
$9.9 < t \leq 10.05$	10	15
$10.05 < t \leq 10.2$	17	32



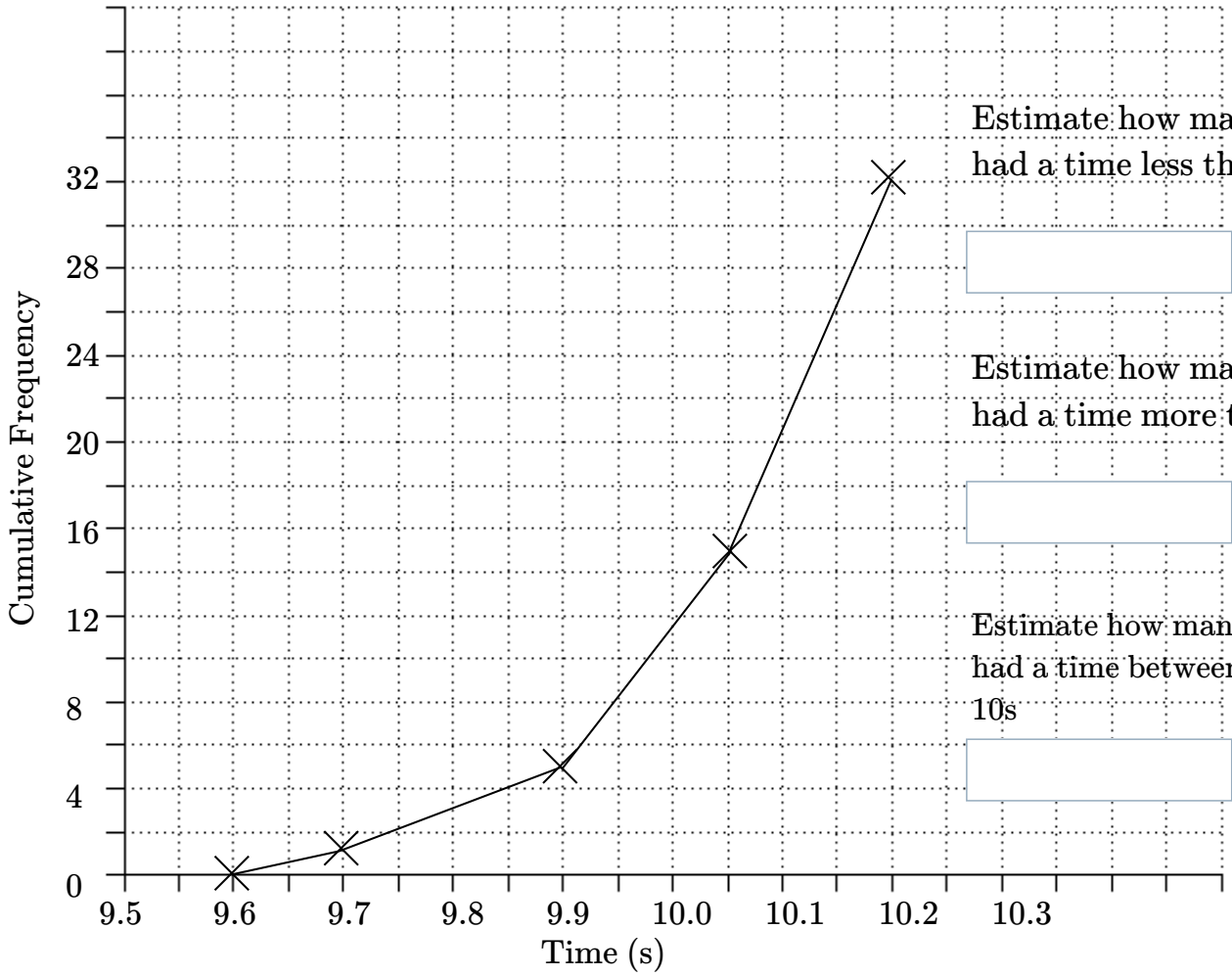
These graphs are intended to show the running total of people/things **up to** a particular value, and are particularly useful in estimating the median and quartiles.

Estimate of Q_2 =

Estimate of Q_1 =

Estimate of Q_3 =

Interquartile Range =



Estimate how many runners had a time less than 10.15s.

runners

Estimate how many runners had a time more than 9.95

runners

Estimate how many runners had a time between 9.8s and 10s

runners

4. A vet records male and female domestic shorthair cat masses in kg.

(a) Draw cumulative frequency diagrams on common axes.

(b) Which group has greater spread?

A female is underweight below 3.2 kg; a male is underweight below 3.8 kg.

(c) Which group has fewer underweight cats?

a

b

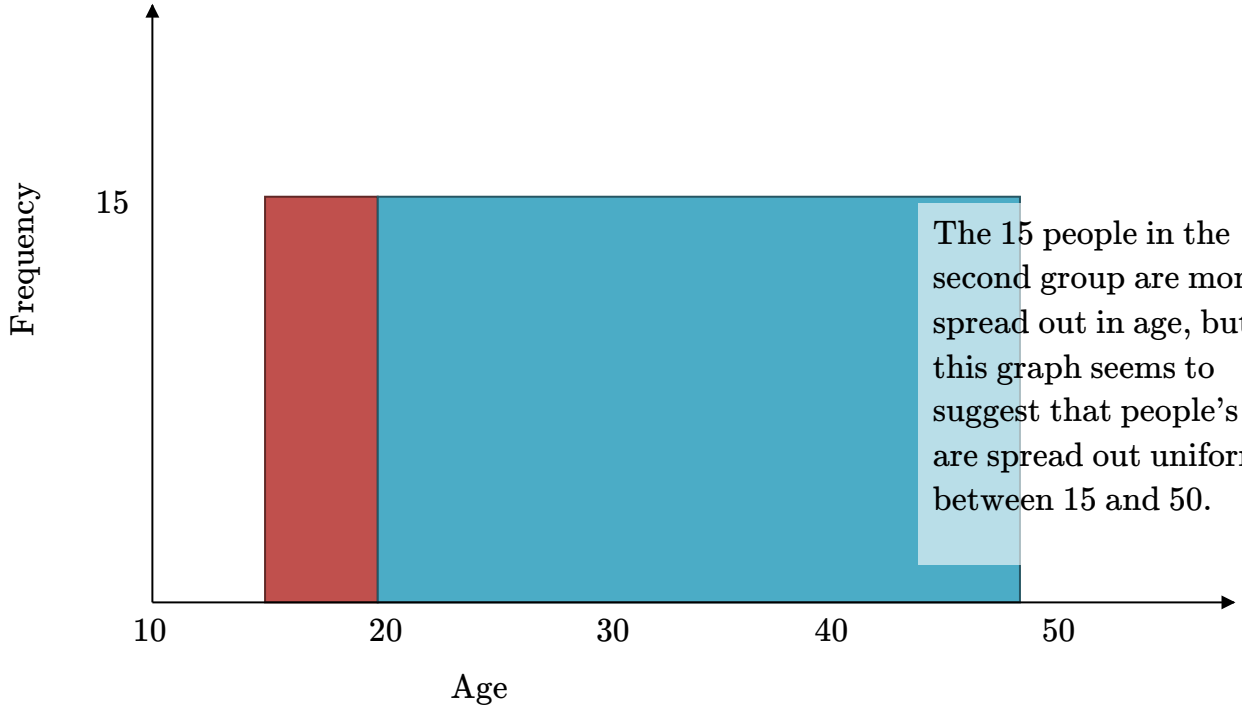
c

Mass (kg)	Male	Female
$2.5 \leq w < 3$	1	5
$3 \leq w < 3.5$	12	17
$3.5 \leq w < 4$	20	32
$4 \leq w < 4.5$	27	12
$4.5 \leq w < 5$	7	4
$5 \leq w < 5.5$	3	0

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Pablo is hosting a party. He counts how many people are between 15 and 20, and 20 and 50.

Why is below graph somewhat unhelpful. How could we fix it?



The 15 people in the second group are more spread out in age, but this graph seems to suggest that people's ages are spread out uniformly between 15 and 50.

Estimating frequency within an age group

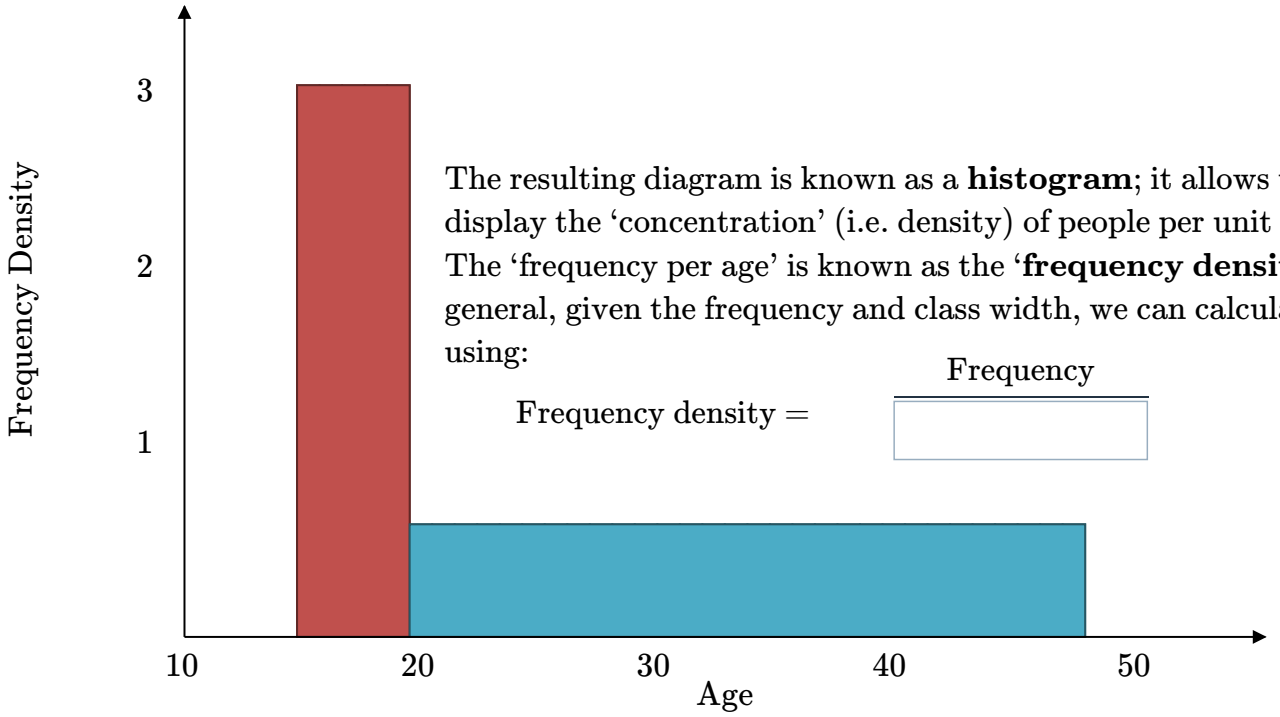
Luisa · 21

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Assume ages are evenly spread within each age group.

[ ] people of each age in the 15–20 group.

[ ] people of each age in the 20–50 group.



The resulting diagram is known as a **histogram**; it allows us to display the 'concentration' (i.e. density) of people per unit value. The 'frequency per age' is known as the '**frequency density**'. In general, given the frequency and class width, we can calculate it using:

$$\text{Frequency density} = \frac{\text{Frequency}}{\square}$$

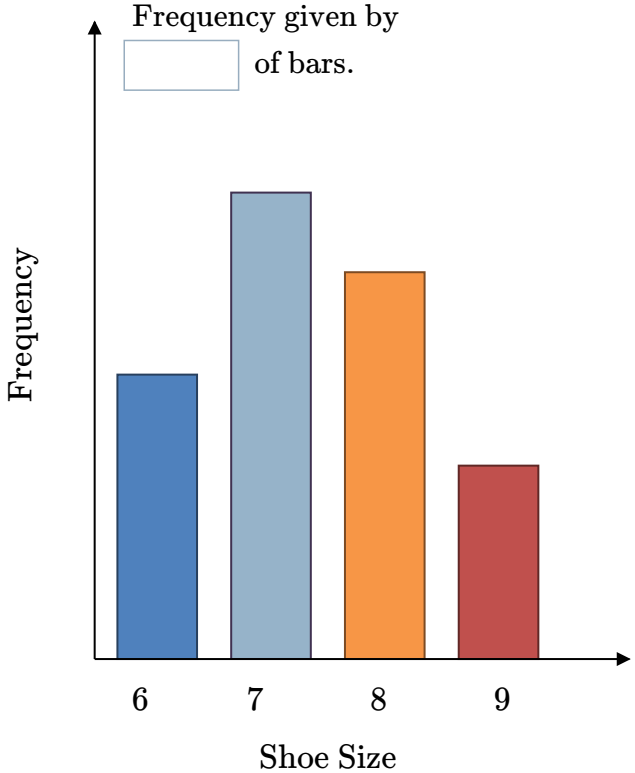
Bar Charts vs Histograms

Luisa · 22

Bar charts

For [ ] data.

Frequency given by [ ] of bars.



Histograms

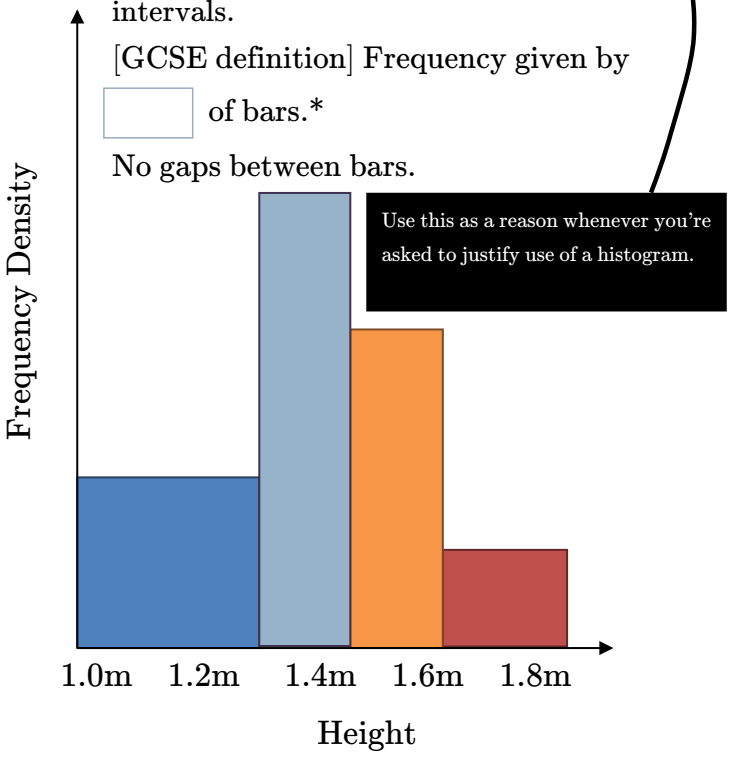
For [ ] data.

Data divided into (potentially uneven) intervals.

[GCSE definition] Frequency given by [ ] of bars.*

No gaps between bars.

Use this as a reason whenever you're asked to justify use of a histogram.



* Not necessarily true. We'll correct this in a sec.

Q1

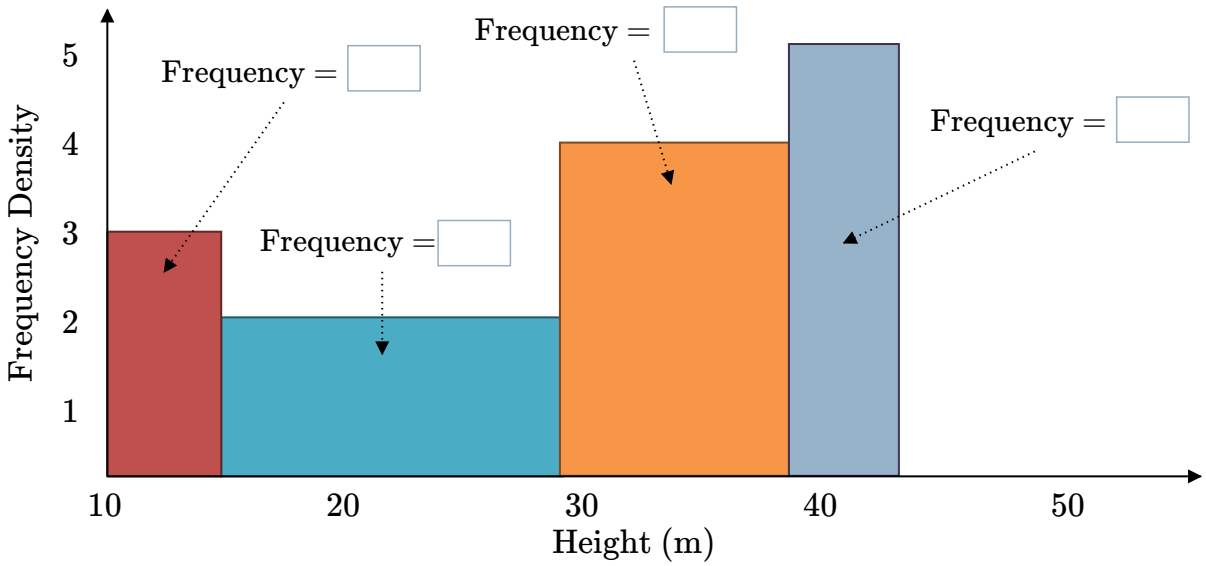
Weight (w kg)	Frequency	Frequency Density
$0 < w \leq 10$	40	
$10 < w \leq 15$	6	
$15 < w \leq 35$		2.6
$35 < w \leq 45$		1

Still using the **incorrect** GCSE formula:

Freq

F.D. Width

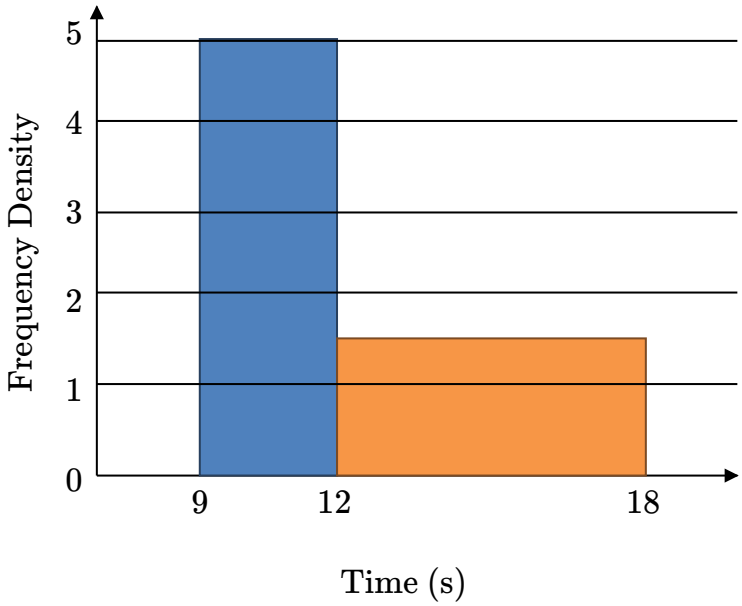
Q2



Unlike at GCSE, the area of a bar is not necessarily equal to the frequency; there are just **proportional**.

✎ Identify the scaling $area \times k frequency$ using a known area with known frequency (which may be total area/frequency or just one bar)

There were 60 runners in a 100m race. The following histogram represents their times. Determine the number of runners with times above 14s.



Total frequency is known; therefore find total area and hence the 'scaling'.

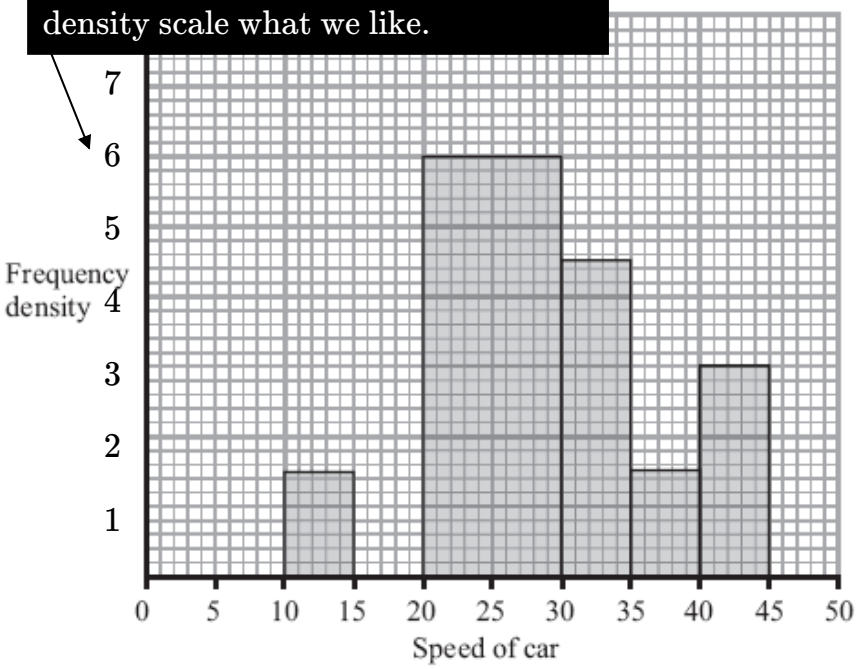
Then use this scaling along with the desired area.

Edexcel S1 May 2012 Q5

A policeman records the speed of the traffic on a busy road with a 30 mph speed limit. He records the speeds of a sample of 450 cars. The histogram in Figure 2 represents the results.

(a) Calculate the number of cars that were exceeding the speed limit by at least 5 mph in the sample. *(4 marks)*

Tip: We can make the frequency density scale what we like.

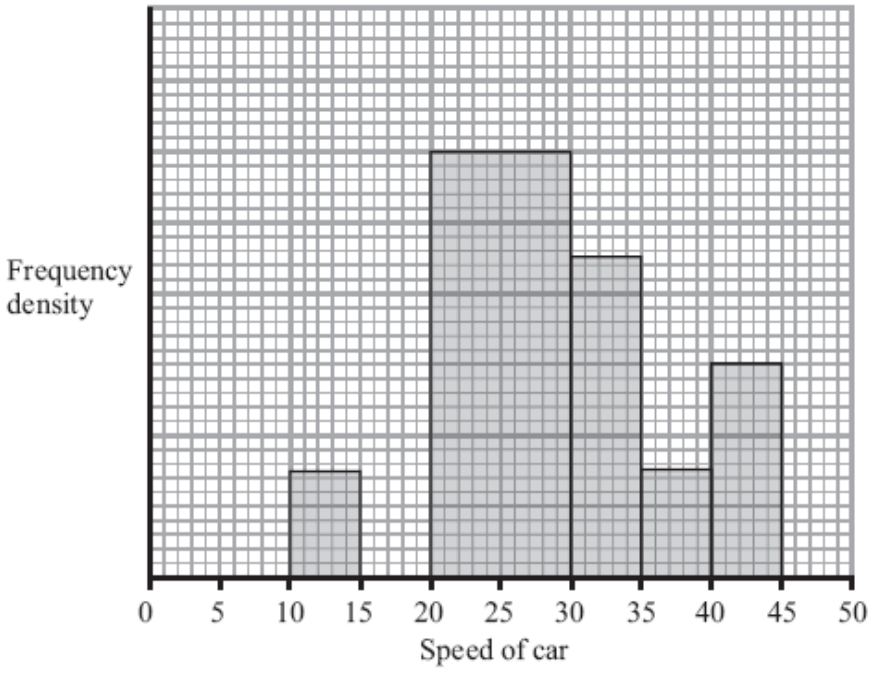


M1 A1: Determine what one small square or one large square is worth.

(i.e. work out $area \rightarrow freq$ scaling)

M1 A1: Use this to find number of cars travelling >35mph.

(b) Estimate the value of the mean speed of the cars in the sample. *(3 marks)*



M1 M1: Use histogram to construct sum of speeds.

A1 Correct value

Tip: Whenever you are asked to calculate mean, median or quartiles from a histogram, form a grouped frequency table. Use your scaling factor to work out the frequency of each bar.

(*c*) Estimate, to 1 decimal place, the value of the median speed of the cars in the sample. **(2)**
(*d*) Comment on the shape of the distribution. Give a reason for your answer. **(2)**
(*e*) State, with a reason, whether the estimate of the mean or the median is a better representation of the average speed of the traffic on the road. **(2)**

(crossed out questions would not appear in new syllabus)

c

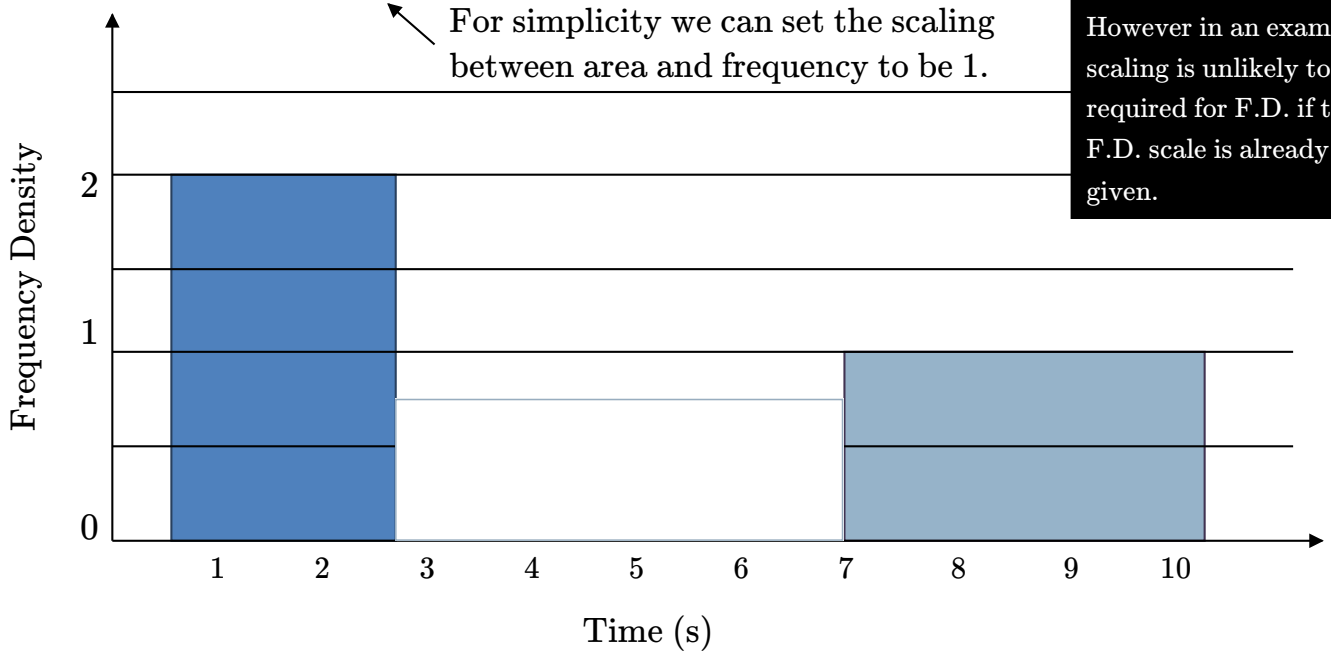
d

e

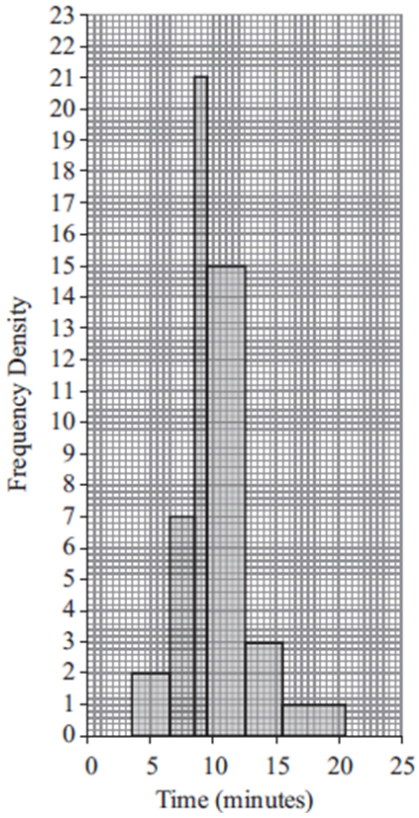
Weight (to nearest kg)	Frequency	F.D.
1-2	4	
3-6	3	
7-9		

For simplicity we can set the scaling between area and frequency to be 1.

Note the gaps affects class width!
Remember the frequency density axis is only correct to scale, so there may be some scaling. However in an exam scaling is unlikely to be required for F.D. if the F.D. scale is already given.



Jan 2012 Q1. The histogram shows the time, to the nearest minute, that a random sample of 100 motorists were delayed by roadworks on a stretch of motorway.



(a) Complete the table. (2 marks)

Delay (minutes)	Number of motorists
4–6	6
7–8	
9	21
10–12	45
13–15	9
16–20	

Tip: Use the correct class widths.

(b) Estimate the number of motorists delayed between 8.5 and 13.5 minutes by the roadworks. (2 marks)

b

An exam favourite is to ask what width and height we'd draw a bar in a drawn histogram.

Q: The frequency table shows some running times. On a histogram the bar for 0-4 seconds is drawn with width 6cm and height 8cm. Find the width and height of the bar for 4-6 seconds.

Time (seconds)	Frequency
$0 \leq t < 4$	8
$4 \leq t < 6$	9

Strategy

Solution

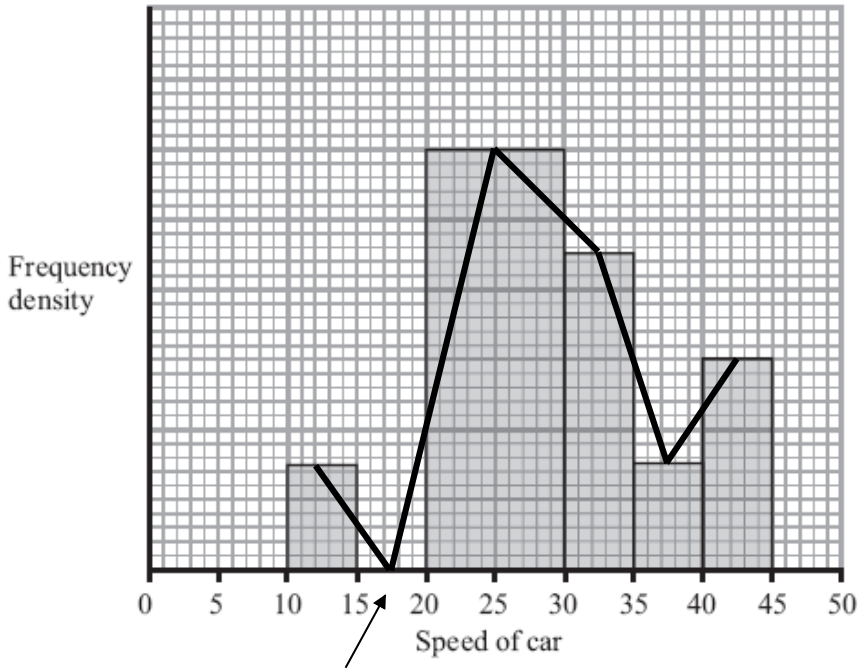
May 2009 Q3. x was measured to the nearest whole number. Forty observations are shown. A histogram bar for 10–15 is 2 cm wide and 5 cm high.
Find (a) the width (1 mark) and (b) the height (2 marks) of the bar for 16–18.

a

b

x	10–15	16–18	19–
Frequency	15	9	16

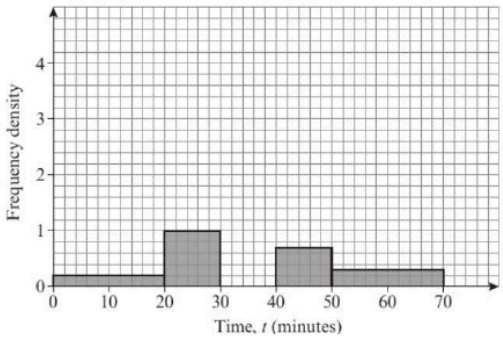
Recall that a frequency polygon can be drawn by using the midpoint of each interval. This corresponds to the midpoint of the top of each bar in a histogram.



Note that the frequency in this interval is 0. That needs to be reflected in the frequency polygon.

5. The partly completed histogram shows airport passenger delays.
(a)(i) Copy and complete the table. (ii) Copy and complete the histogram. (4 marks)
(b) Estimate how many passengers were delayed between 25 and 38 minutes. (2 marks)

Time (min)	Frequency
$0 \leq t < 20$	4
$20 \leq t < 30$	
$30 \leq t < 35$	15
$35 \leq t < 40$	25
$40 \leq t < 50$	
$50 \leq t < 70$	



a(i)

a(ii)

b

6. The variable y was measured to the nearest whole number. 60 observations were taken and are recorded in the table below.

y	10–12	13–14	15–17	18–25
Frequency	6	24	18	12

(a) Write down the class boundaries for the 13–14 class. (1 mark)

A histogram was drawn and the bar representing the 13–14 class had a width of 4 cm and a height of 6 cm. For the bar representing the 15–17 class, find:
(i) the width (1 mark)
(ii) the height (2 marks).

Problem-solving: Remember that area is proportional to frequency.

a

b(i)

b(ii)

Exercise 3D – Statistics Year 1 Pages 50-52 **Luisa · 35**

7. Daily mean temperatures in Leeming in May 2015 are summarised below (large data set; © Crown Copyright Met Office). The $8 \le t < 10$ bar is 1 cm wide and 8 cm high.

(a) Find the width and height of the $11 \le t < 12$ bar. (2 marks)

(b) Estimate the mean and standard deviation. (3 marks)

(c) Interpolate the lower quartile. (2 marks)

(d) Estimate the number of days exceeding the mean plus one standard deviation. (2 marks)

Temperature (°C)	Frequency
$4 \le t < 8$	4
$8 \le t < 10$	8
$10 \le t < 11$	6
$11 \le t < 12$	7
$12 \le t < 15$	5
$15 \le t < 16$	1

a

b

c

d

Exercise 3E – Statistics Year 1 Pages 54 **Luisa · 36**

4. The table gives sampled daily maximum relative humidities from the large data set for Leuchars and Camborne during 2015. © Crown Copyright Met Office.

Leuchars	100	98	100	100	100	100	100	100	94	100	91	100	100	89	100
Camborne	92	95	99	96	100	100	90	98	81	99	100	99	91	98	100

(a) Find the median and quartiles for both samples. (4 marks)

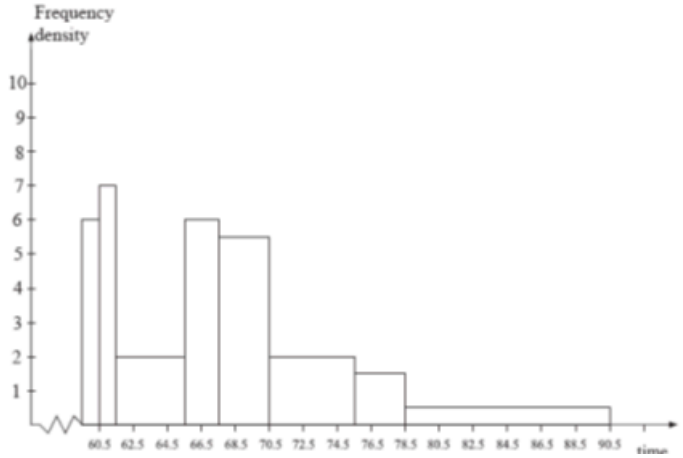
(b) Compare the samples. (2 marks)

a

b

Supplementary Exercise — Luisa · 37

Jan 2008 Q3. The histogram shows time, measured to the nearest minute, for 140 runners to complete a fun run. Calculate how many took between 78.5 and 90.5 minutes. (5 marks)



3

Supplementary Exercise — Luisa · 38

2. The distances, to the nearest km, travelled by 134 examiners to a London meeting are summarised below.

(a) Justify the use of a histogram. (1 mark)

(b) Calculate the frequency densities needed. Do not draw the histogram. (2 marks)

Distance (km)	Examiners
41–45	4
46–50	19
51–60	53
61–70	37
71–90	15
91–150	6

a

b

Supplementary Exercise — Luisa · 39

May 2013 (R) Q3. Yields of 70 tomato plants are given in the table. Use $\sum fx = 755$, $\sum fx^2 = 12037.5$. The $5 \le y < 10$ bar is 1.5 cm wide, 8 cm high.

(a) Find the width and height for $15 \le y < 25$. (3 marks)
(b) Interpolate the median. (2 marks)
(c) Estimate the mean and standard deviation. (4 marks)
(d) Describe and justify skewness. (2 marks)

Yield (kg)	Frequency	Midpoint
$0 \le y < 5$	16	2.5
$5 \le y < 10$	24	7.5
$10 \le y < 15$	14	12.5
$15 \le y < 25$	12	20
$25 \le y < 35$	4	30

a

b

c

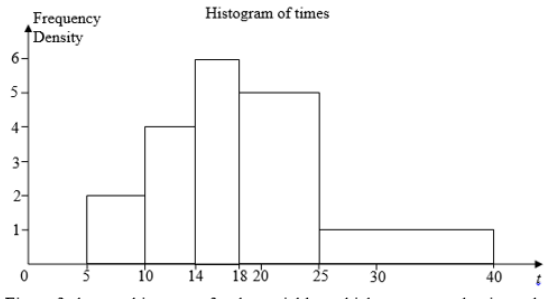
d

Supplementary Exercise — Luisa · 40

Q4. June 2007 Q5. The histogram shows times t (minutes) to swim 500 m.

(a) Complete the frequency table. (2 marks)
(b) Estimate how many took longer than 20 minutes. (2 marks)
(c) Estimate the mean. (4 marks)
(d) Estimate the standard deviation. (3 marks)
(e) Find the median and quartiles. (4 marks)
(f) Calculate $3(\text{mean} - \text{median})/\text{SD}$ and describe skewness. (2 marks)

Time	5–10	10–14	14–18	18–25	25–40
Frequency	10	16	24		



a

b

c

d

e

f

Supplementary Exercise — Luisa · 41

Jan 2013 Q5. The table summarises weekly incomes of 100 households. Use $\sum fy^2 = 12452800$. The $200 \le y < 240$ histogram bar is 2 cm wide and 7 cm high.

(a) Find width and height for $320 \le y < 400$. (3 marks)

(b) Interpolate the median income to the nearest pound. (2 marks)

(c) Estimate the mean and standard deviation. (4 marks)

(d) Use $3(\text{mean} - \text{median})/\text{SD}$ to calculate and describe skewness. (2 marks)

Income (£)	Midpoint	Frequency
$0 \le y < 200$	100	12
$200 \le y < 240$	220	28
$240 \le y < 320$	280	22
$320 \le y < 400$	360	18
$400 \le y < 600$	500	12
$600 \le y < 800$	700	8

a

b

c

d

Supplementary Exercise — Luisa · 42

May 2010 Q5. A random sample of 56 students recorded weekly television viewing to the nearest hour.

(a) Find the missing midpoints. (2 marks)

(b) The 11–20 bar measures 4 cm by 6 cm. Find the width and height for 26–30. (3 marks)

(c) Estimate the mean and SD. (5 marks)

(d) Interpolate the median. (2 marks)

(e) With $Q_1 = 15.8$ and $Q_3 = 29.3$, describe skewness and give a reason. (2 marks)

Hours	1–10	11–20	21–25	26–30	31–40	41–59
Frequency	6	15	11	13	8	3
Midpoint	5.5	15.5		28		50

a

b

c

d

e

Jan 2009 Q5. 104 teenagers reported time spent shopping last month, to the nearest hour. The 8–10 histogram bar was 1.5 cm wide and 3 cm high.

(a) Find the width and height for 16–25 hours. (3 marks)

(b) Interpolate the median and IQR. (5 marks)

(c) Estimate the mean and SD. (4 marks)

(d) Describe skewness with a reason. (2 marks)

(e) Recommend an average and measure of spread with a reason. (2 marks)

Hours	Midpoint	Frequency
0–5	2.75	20
6–7	6.5	16
8–10	9	18
11–15	13	25
16–25	20.5	15
26–50	38	10

a

b

c

d

e